DSAI4
Module 2 Project

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Olist Data Team

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01 **Executive Summary**
Overview of Olist's strategic direction

02 **Business Value Proposition**
Strategic impact on service & valuation

03 **Warehousing Strategy**
Distribution center & logistics analysis

04 **System Architecture**
Meltano, dbt, and Dagster pipeline

05 **Technical Solution Overview**
Medallion architecture & BigQuery

06 **Risk & Mitigation**
Data quality & implementation patches

Q&A Session

Executive Summary

Company Profile

Hyper-growth Brazilian E-Commerce leader focused on scaling operations.

Current Challenge

Existing rental warehousing solutions no longer support rapid business expansion needs.
Logistic efficiency to ensure product is delivered to the customer with most cost effective and faster way

Strategic Initiatives & Objectives

- Establish self-owned regional warehouses in major Brazilian cities to enhance asset valuation.
- Execute comprehensive cost evaluation for proprietary product storage infrastructure.
- Optimize logistics efficiency as a secondary strategic objective to maintain service excellence.

Business Value Proposition: Strategic Impact

Service & Delivery Optimization

Maintain customer service via better delivery timings

Target: Moving Customer Buckets

- **At Risk (24%):** Target <20% within 3 months of new warehouse opening
- **Potential Loyalists (31%):** Goal >34% via transition from At Risk
- **Loyal Customers (23%):** Strategy: Maintain

Asset Appreciation

Increase valuation from owned warehouse/land

20% increase on financial statements for warehouse/land assets

Operational Efficiency

Reduced reliance on rental facilities

30% decrease in total rental costs

Key Business Analysis Findings

Revenue Trend

Consistent growth (2017–2018); Nov 2016 data gap noted.

Peak Order Months

May(Mother's Day), July (Peak Sch Break), Aug(Father's Day) peak volume; monthly average ~8,600 orders.

Top Category

Bed Bath Table holds the highest order count across all years.

Cancellation Rate

<1% rate indicates very high platform reliability.

Shipping vs Review

Negative correlation: faster delivery directly leads to higher scores.

Top State (Revenue)

São Paulo dominates across all product categories.

Largest Customer Segment

Potential Loyalists: This segment represents the largest opportunity for conversion into Champions or Loyal customers.

Strategic Warehousing & Revenue Distribution

Geographic footprint of revenue-generating states supporting the Sao Paulo Central Hub

Active States

- **Sao Paulo** (Central Hub)
- **Parana**
- **Minas Gerais**
- **Rio de Janeiro**
- **Santa Catarina**
- **Rio Grande do Sul**

Note: Sao Paulo dominates across all product categories

Warehousing Strategy: Centralized Hub (Solution #1)

Primary Node: Sao Paulo
Central Distribution Warehouse

Computer Accessories
Distribution to: Parana, Minas Gerais

Sports Leisure
Distribution to: Parana, Rio de Janeiro, Santa Catarina

Furniture Decor
Distribution to: Parana, Minas Gerais

Health Beauty
Distribution to: Parana, Minas Gerais, Rio de Janeiro, Santa Catarina, Rio Grande do Sul

Warehousing Strategy: Decentralized Distribution (Solution #2)

No Central Distribution Center
Direct regional fulfillment strategy

Computer Accessories
Warehouses: Sao Paulo, Parana, Minas Gerais

Sports Leisure
Warehouses: Sao Paulo, Parana, Rio de Janeiro, Santa Catarina

Furniture Decor
Warehouses: Sao Paulo, Parana, Minas Gerais

Health Beauty
Warehouses: Sao Paulo, Parana, Minas Gerais, Rio de Janeiro, Santa Catarina, Rio Grande do Sul

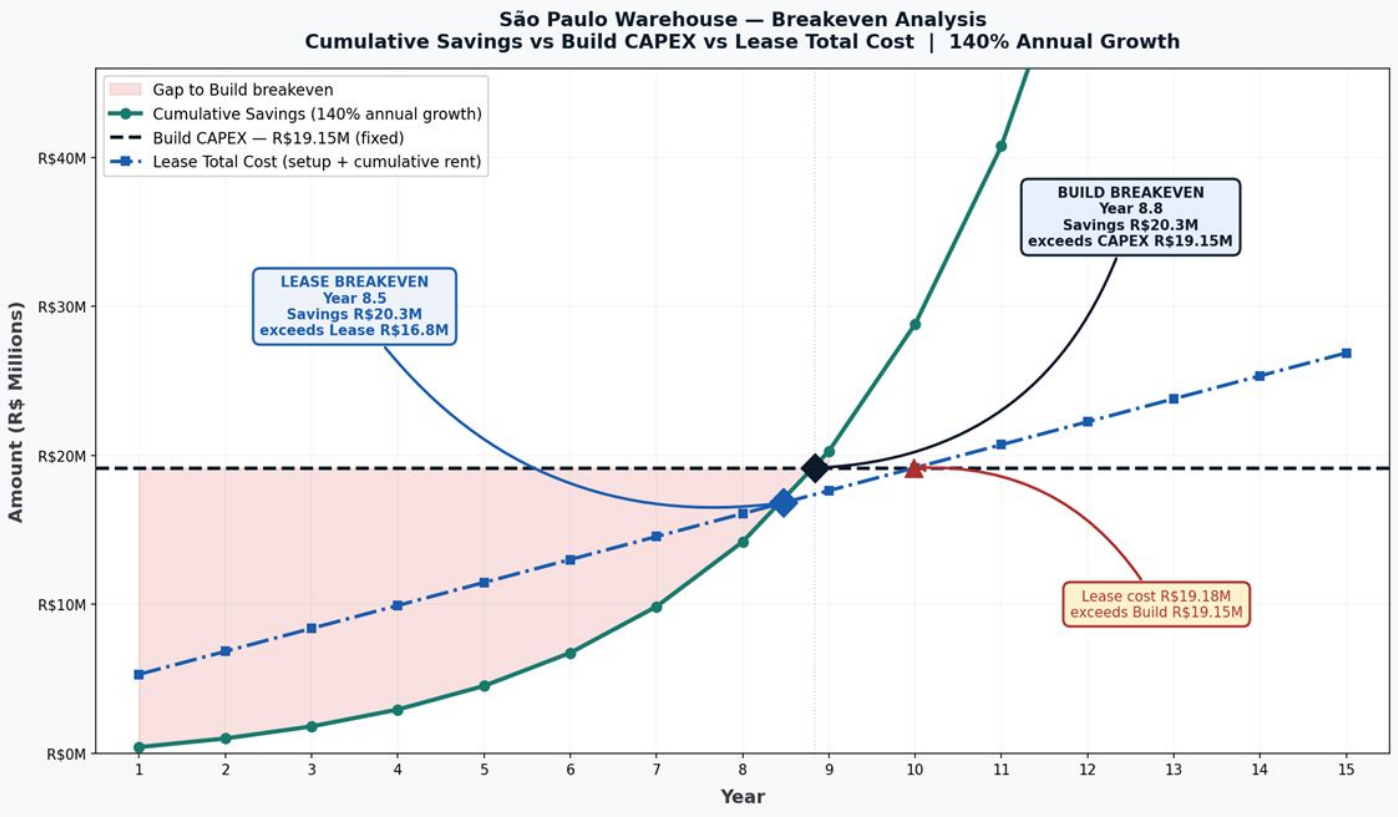
Investment Analysis: Option A — Build (Own) Warehouse

Cost Component (CAPEX)	Sao Paulo	Minas Gerais	Santa Catarina
Land Purchase	R$ 4,500,000	R$ 2,250,000	R$ 1,750,000
Construction	R$ 11,000,000	R$ 9,750,000	R$ 9,000,000
Racking Equipment	R$ 2,500,000	R$ 2,200,000	R$ 2,000,000
IT & Warehouse System	R$ 1,150,000	R$ 1,050,000	R$ 1,000,000
TOTAL CAPEX	R$ 19,150,000	R$ 15,250,000	R$ 13,750,000
ROI			
5-Year Total Savings	R$ 2,064,315	R$ 623,470	R$ 414,500
5-Year ROI — current volume	−89.2%	−95.9%	−97.0%
5-Year ROI — at 140% growth	−76.4%	−91.1%	−93.4%
9-Year Cumulative Savings	R$ 20,293,297	R$ 6,129,036	R$ 4,074,752
9-Year ROI — at 140% growth	+6.0%	−59.8%	−70.4%
Breakeven year	**Year 9 ✓**	**Year 12 ✓**	**Year 13 ✓**

Investment Analysis: Option B — Lease (Rent) Warehouse

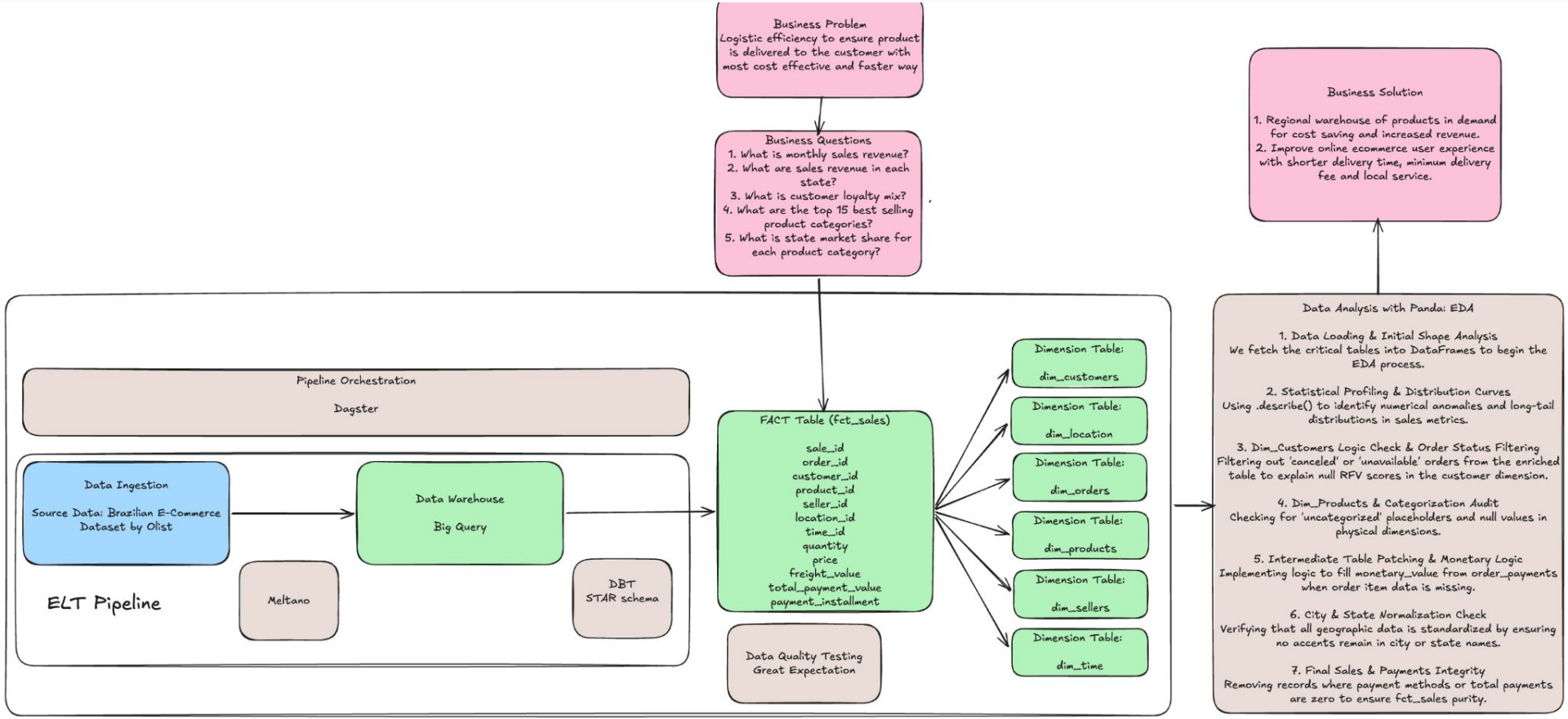
Cost Component (LEASE)	Sao Paulo	Minas Gerais	Santa Catarina
Setup Cost	R$ 3,755,000	R$ 3,345,000	R$ 3,097,500
Annual Rent	R$ 1,542,000	R$ 1,140,000	R$ 990,000
5-Year Rent	R$ 7,710,000	R$ 5,700,000	R$ 4,950,000
LEASE 5-YEAR TOTAL COST	R$ 11,465,000	R$ 9,045,000	R$ 8,047,500
ROI			
5-Year Total Savings	R$ 2,064,315	R$ 623,470	R$ 414,500
5-Year ROI — current volume	−82.0%	−93.1%	−94.8%
5-Year ROI — at 140% growth	−60.6%	−84.9%	−88.7%
9-Year Cumulative Benefit	R$ 20,293,297	R$ 6,129,036	R$ 4,074,752
Lease Total Cost at Year 9	R$ 17,633,000	R$ 13,605,000	R$ 12,007,500
Break-even year	**Year 9 ✓**	**Year 12 ✓**	**Year 13 ✓**

Breakeven Analysis

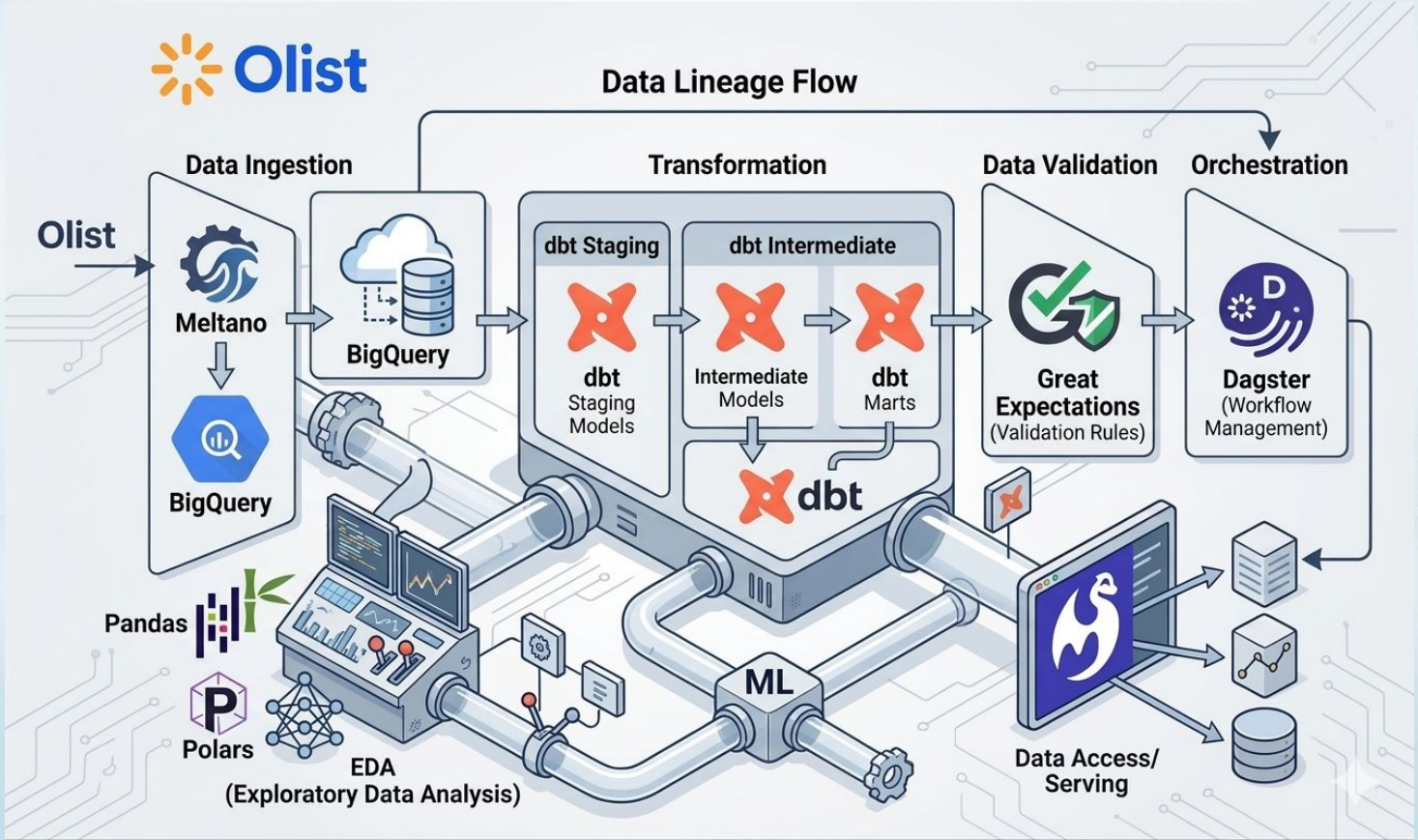


System Architecture & Technical Solution Overview

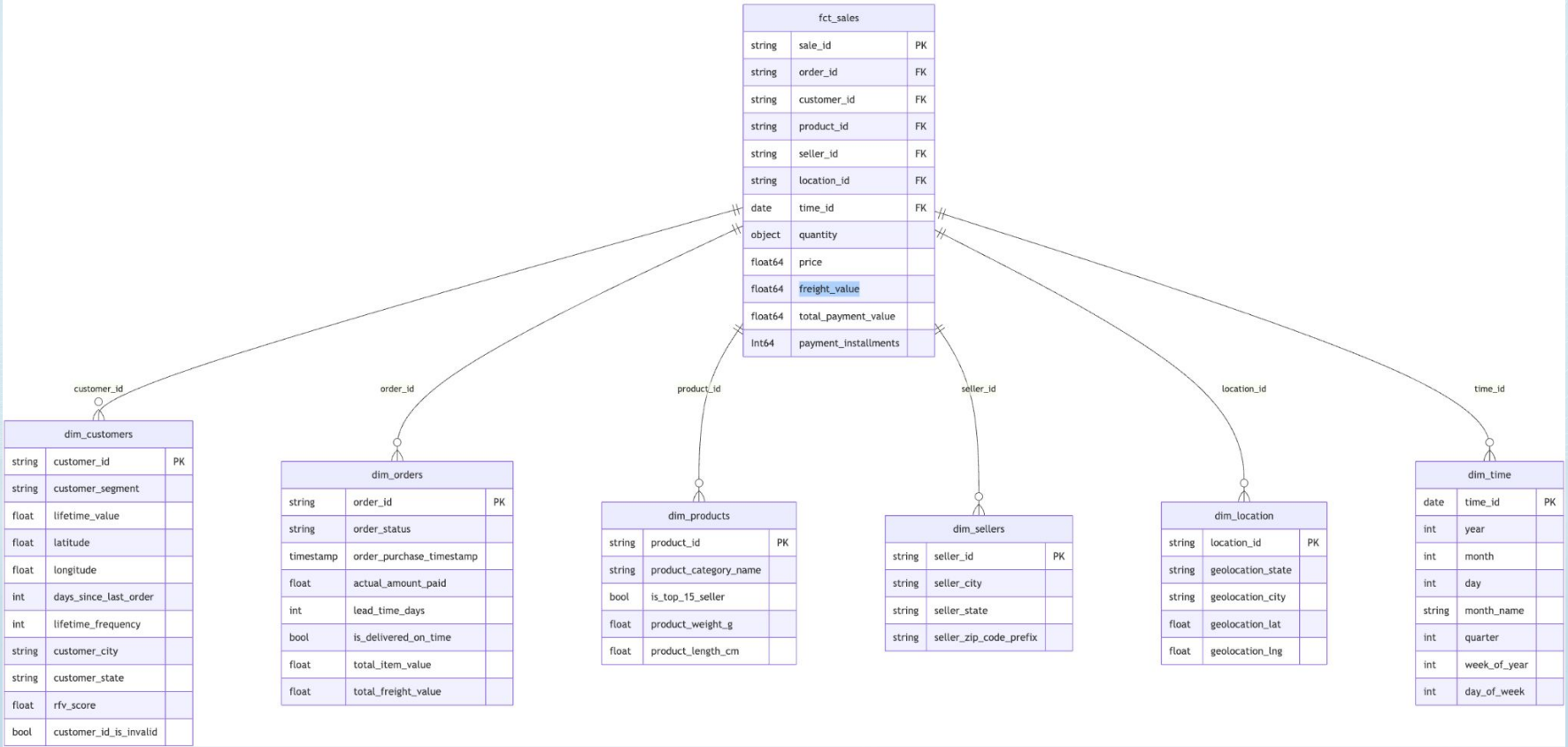
System Architecture Overview



Technical Solution Overview



Technical Solution Overview



Core Components

meltano/	Meltano	Data Extraction & Loading (EL) — ingests raw Olist CSV datasets into BigQuery via configured tap/target plugins
dagster/	Dagster	Orchestration & Workflow Management — defines asset lineage, schedules, quality gates, and auto-generates dbt docs
dbt_olist/models/staging/	dbt	Raw source models as views — light renaming, type casting, and deduplication of source tables
dbt_olist/models/intermediate/	dbt	Business logic layer — RFV segmentation, order enrichment, customer metrics, product categorization
dbt_olist/models/marts/core/	dbt	Fact & Dimension tables for Olist sales — fct_sales, dim_customers, dim_orders, dim_products, dim_sellers, dim_location, dim_time
eda/eda.ipynb	Jupyter + Panda	Exploratory data analysis — statistical profiling, null checks, geographic normalization audits, and data cleaning validation
salesportal.py / .streamlit/	Streamlit + Plotly	Executive dashboard — KPIs, monthly revenue trends, state-level sales, RFM segmentation, and product category analysis

Risk & Mitigation

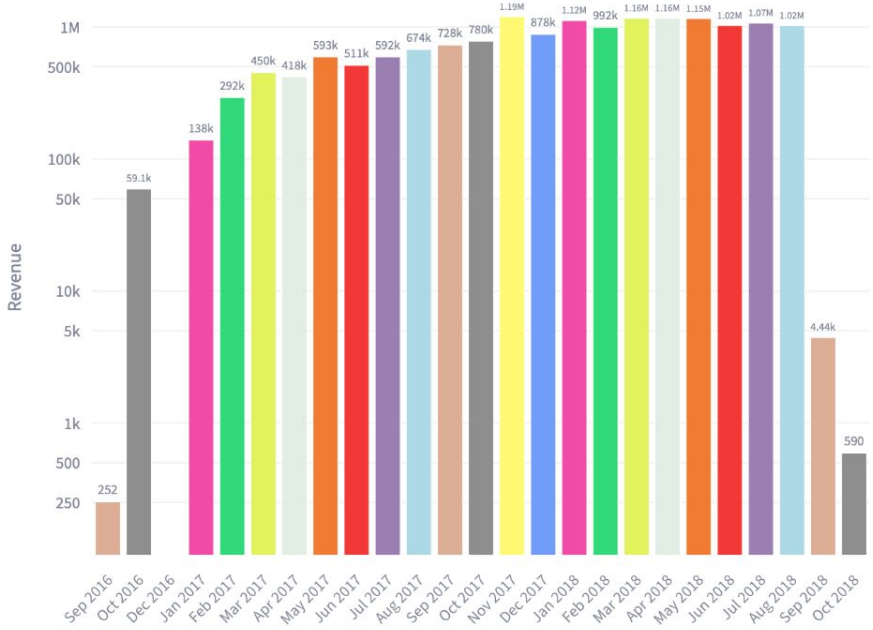
Risk & Mitigation

- 🔍 Manual Fixes
  - 📍 Geolocation Patching (Seed)
  - 👤 dim_customers — RFV Null & Segment Fixes
- 🛠️ Technical Implementation & Manual Patches
  - 📦 Product Dimension & Categorization
  - 📐 Loyalty Logic (RFV) & Frequency Repair
  - 🔢 Float Precision & Schema Standardization
  - 📍 Geolocation Standardization
  - 🔗 Table Join Key Discrepancy

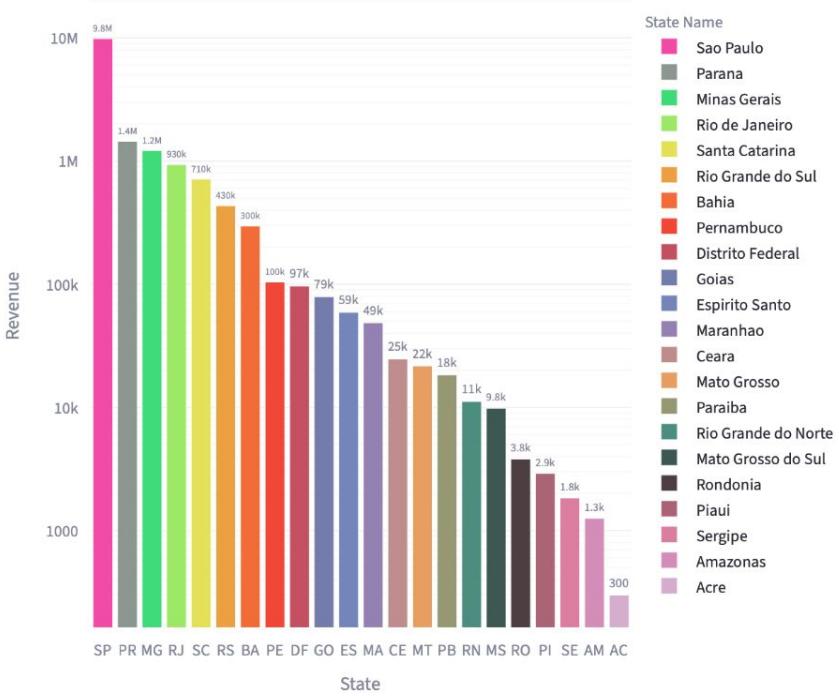
Dashboard (Demo)

Customers	New Prospects	Revenue	Order Volume	Products Sold	Average Order Value
96.1K	**1,397**	**R$ 16.01M**	**99.44K**	**113.42K**	**R$ 160.99**

Monthly Sales Revenue

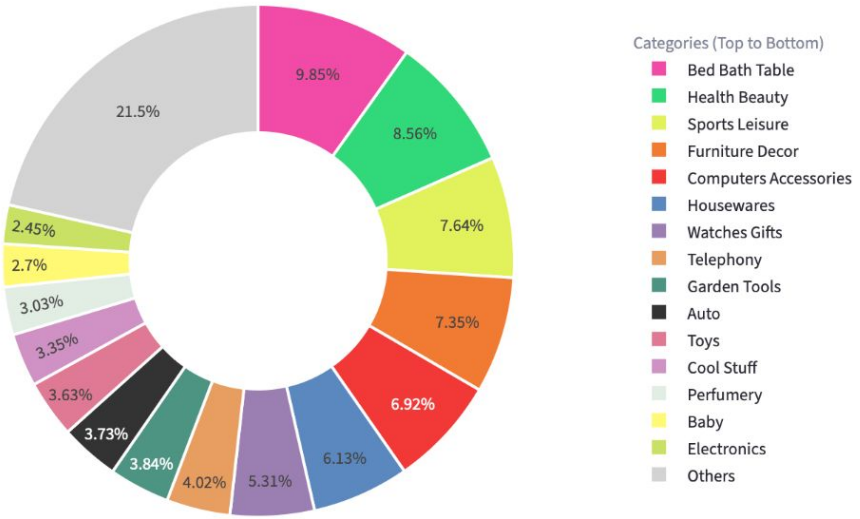


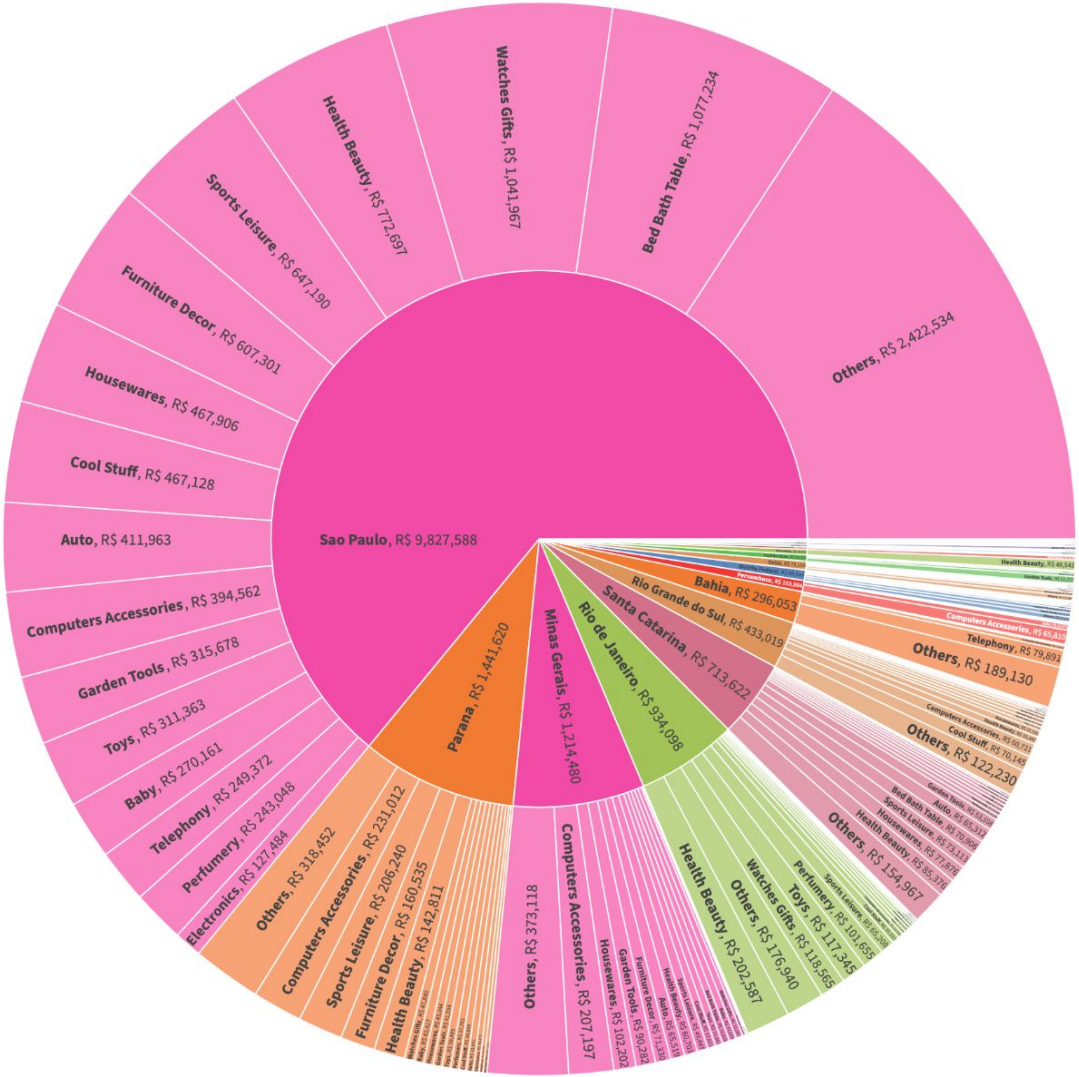
Sales Revenue by States



Customer Loyalty Mix

Top 15 Best Selling Product Categories




Sao Paulo, R$ 9,827,588
Parana, R$ 1,441,620
Minas Gerais, R$ 1,214,480
Rio de Janeiro, R$ 934,098
Santa Catarina, R$ 713,622
Rio Grande do Sul, R$ 433,019
Bahia, R$ 296,053
Others, R$ 2,422,534
Bed Bath Table, R$ 1,077,234
Watches Gifts, R$ 1,041,967
Health Beauty, R$ 772,697
Sports Leisure, R$ 647,190
Furniture Decor, R$ 607,301
Housewares, R$ 467,906
Cool Stuff, R$ 467,128
Auto, R$ 411,963
Computers Accessories, R$ 394,562
Garden Tools, R$ 315,678
Toys, R$ 311,363
Baby, R$ 270,161
Telephony, R$ 249,372
Perfumery, R$ 243,048
Electronics, R$ 127,484
Others, R$ 318,452
Computers Accessories, R$ 231,012
Sports Leisure, R$ 206,240
Furniture Decor, R$ 160,535
Health Beauty, R$ 142,811
Others, R$ 373,118
Computers Accessories, R$ 207,197
Housewares, R$ 102,202
Garden Tools, R$ 90,285
Health Beauty, R$ 202,587
Others, R$ 176,940
Watches Gifts, R$ 118,565
Toys, R$ 117,345
Perfumery, R$ 101,655
Others, R$ 154,967
Health Beauty, R$ 85,376
Housewares, R$ 73,143
Sports Leisure, R$ 77,476
Bed Bath Table, R$ 75,908
Auto, R$ 66,312
Garden Tools, R$ 64,610
Cool Stuff, R$ 70,146
Others, R$ 122,230
Computers Accessories, R$ 65,731
Telephony, R$ 79,891
Computers Accessories, R$ 65,810
Health Beauty, R$ 48,543
State Name
Acre
Amazonas
Bahia
Ceara
Distrito Federal
Espirito Santo
Goias
Maranhao
Mato Grosso
Mato Grosso do Sul
Minas Gerais
Paraiba
Parana
Pernambuco
Piaui
Rio Grande do Norte
Rio Grande do Sul
Rio de Janeiro
Rondonia
Santa Catarina
Sao Paulo
Sergipe

Questions & Answer

Appendix

System Architecture & Documentation

Visual Pipeline Documentation

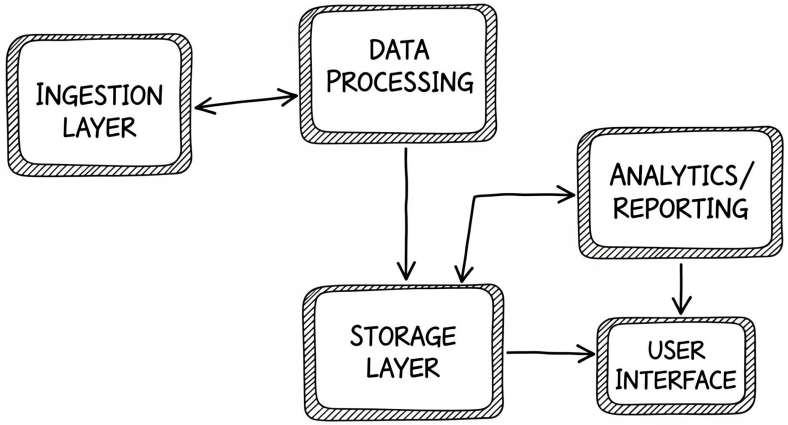
Utilizing **Excalidraw** for high-fidelity architectural mapping:

- End-to-end data lineage visualization
- Component-level interaction diagrams
- Collaborative design for stakeholder review

Technical Clarity

Translating complex ELT logic into executive-friendly visuals:

- Meltano to BigQuery ingestion flows
- dbt Medallion layer transformations
- Dagster orchestration checkpoints



Conceptual System Diagram via Excalidraw Interface

Business Value Proposition: Strategic Impact

Service Excellence & Delivery

By owning local warehouses, we eliminate dependency on third-party schedules, directly maintaining high customer service standards via faster last-mile fulfillment and better delivery timings.

ASSET VALUATION

20% ↑

Projected Equity Increase

RENTAL COSTS

30% ↓

Reduced Dependency

Business Value Proposition: Strategic Impact

AT RISK (Target <20%)

24% Current

3 months post-warehouse opening.

POTENTIAL LOYALISTS

> 34% Target

Conversion increase from At Risk.

LOYAL CUSTOMERS

23% Baseline

Maintain current high-value base.

Strategic Real Estate Equity

Transitioning from rental to ownership increases total asset valuation on financial statements through warehouse and land equity.

Projected Financial Impact

20% Increase

in asset valuation on financial statements.

Rental Reliance Reduction

30% Decrease

in overall rental costs annually.

Executive Summary

- We are a hyper growth Brazilian E-Commerce company
- Our current rental warehousing solution no longer fits our purpose
- We are looking to build our own regional warehouses in major Brazilian cities
- Evaluating the cost to build the warehouses for storage of our products
- Need to consider secondary objective of efficient logistics

Business Va

- Maintain cu
- Increase as
- Less reliant

📉 Business Analysis Findings

Business Question	Finding
Revenue trend over time	Consistent growth from 2017–2018; Nov 2016 missing entirely, Dec 2016 has 1 order only — known Kaggle source gap
Peak order months	August, May, July are highest volume months; monthly average ~8,600 orders
Top product category	`bed_bath_table` — highest order count across all years
Highest AOV category	`computers` — customers spend more per order despite lower frequency
Dominant payment method	Credit card (~74%), followed by boleto (~19%)
Repeat purchase rate	~3% of customers return — loyalty programs critical to improve retention
Review score vs revenue	Positive correlation — score 5 drives highest orders and revenue
Review score vs AOV	Negative correlation — low-rated products tend to be expensive items
Cancellation rate	<1% — high platform reliability confirmed
Shipping vs review score	Negative correlation — faster delivery = higher review score
Top state by revenue	São Paulo dominates all product categories
Largest customer segment	Potential Loyalists — biggest opportunity for conversion to Loyal/Champions

📉 Business Analysis Findings

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Warehousing Strategy (#1)

- Central Distribution Warehouse in Sao Paulo
  - Computer Accessories: Parana, Minas Gerais
  - Sports Leisure: Parana, Rio de Janeiro, Santa Catarina
  - Furniture Decor: Parana, Minas Gerais
  - Health Beauty: Parana, Minas Gerais, Rio de Janeiro, Santa Catarina, Rio Grande do Sul

Warehousing Strategy (#2)

- No Central Distribution Center
  - Computer Accessories: Sao Paulo, Parana, Minas Gerais
  - Sports Leisure: Sao Paulo, Parana, Rio de Janeiro, Santa Catarina
  - Furniture Decor: Sao Paulo, Parana, Minas Gerais
  - Health Beauty: Sao Paulo, Parana, Minas Gerais, Rio de Janeiro, Santa Catarina, Rio Grande do Sul

Business Value Proposition (End)

- Maintain customer service via better delivery timings
  - Moving the customer buckets
    - At Risk (24%): Target of <20%, 3 months after new warehouse opening
    - Potential Loyalists (31%): > 34%, increase from At Risk
    - Loyal Customers (23%): Maintain
- Increase assets valuation from having own warehouse / land
  - Warehouse / land assets: 20% increase on financials statements
- Less reliant on rental warehouse
  - Decrease of 30% on rental cost

Boa Vista
RORAIMA
Oiapoque
AMAPA
Macapa
Belem
AMAZONAS
Manaus
Careiro
Benjamin Constant
Santarem
Altamira
Traituba
PARA
Maraba
Carajas
Araguaina
Cachimbo
Sao Luis
Parnaiba
MARANHAO
Teresina
CEARA
Fortaleza
RIO GRANDE DO NORTE
Natal
Joao Pessoa
PARAIBA
Picos
PIAUI
Salgueiro
PERNAMBUCO
Recife
Juazeiro
ALAGOAS
Maceio
Aracaju
SERGIPE
Cruzeiro do Sul
Boca do Arca
Humaita
ACRE
Rio Branco
Assis Brasil
Porto Velho
Guajara-Mirim
RONDONIA
Palmas
TOCANTINS
Barreiras
BAHIA
Salvador
Ilheus
Vitoria da Conquista
MATO GROSSO
Cuiaba
Caceres
Rondonopolis
GOIAS
Brasilia
DISTRITO FEDERAL
Goiania
MINAS GERAIS
Uberlandia
Belo Horizonta
ESPIRITO SANTO
Vitoria
MATO GROSSO DO SUL
Corumba
Santa Fe do Sul
Campa Grande
Panorama
SAO PAULO
Ponta Pora
RIO DE JANEIRO
Rio de Janeiro
Sao Paulo
Santos
PARANA
Foz do Iguacu
Curitiba
Sao Francisco do Sul
SANTA CATARINA
Florianopolis
RIO GRANDE DO SUL
Santa Maria
Porto Alegre
Rio Grande

Cachimbo
Salgueiro
PERNAMBU
Juazeiro
Palmas
TOCANTINS
Barreiras
BAHIA
Ara
SERG
MATO
GROSSO
Salvador
GOIAS
Cuiaba
Brasilia
Ilheus
Vitoria da Conquista
Caceres
Rondonopolis
Goiania
DISTRITO
FEDERAL
MINAS
GERAIS
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Santa Fe
do Sul
Uberlandia
Belo
Horizonta
ESPIRITO
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